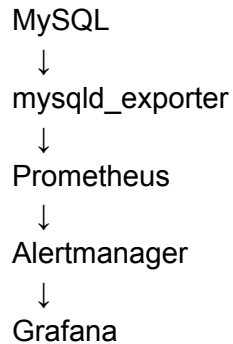


Prometheus, Grafana, MySQL Exporter & Alertmanager Notes

Monitoring Architecture



Components

MySQL

- Stores application data.
- Generates database metrics.

mysqld_exporter

- Collects MySQL metrics.
- Exposes metrics in Prometheus format.
- Runs on port 9104.

Prometheus

- Scrapes metrics from exporters.
- Stores time-series data.
- Evaluates alert rules.
- Runs on port 9090.

Alertmanager

- Receives alerts from Prometheus.
- Handles grouping, silencing, and routing.
- Sends notifications through Email, Slack, Telegram, etc.

- Runs on port 9093.

Grafana

- Visualizes metrics.
 - Creates dashboards and panels.
 - Uses Prometheus as a data source.
 - Runs on port 3000.
-

Prometheus Configuration

Global Configuration

global:

scrape_interval: 5s

evaluation_interval: 5s

scrape_interval

- Frequency of metric collection.

evaluation_interval

- Frequency of alert rule evaluation.
-

Rule Files

rule_files:

- alerts.yml

Purpose:

- Loads alert definitions.
-

Alertmanager Integration

alerting:

alertmanagers:

- static_configs:
- targets:
 - alertmanager:9093

Purpose:

- Sends firing alerts to Alertmanager.
-

Scrape Configurations

Prometheus Self Monitoring

- job_name: 'prometheus'
- static_configs:
 - targets:
 - prometheus:9090

MySQL Exporter Monitoring

- job_name: 'mysql_exporter'
 - static_configs:
 - targets:
 - mysql_exporter:9104
 - labels:
 - env: production
 - database: primary
-

MySQL Exporter

Purpose

Collects metrics such as:

- Active connections
- Queries

- Slow queries
 - InnoDB statistics
 - Replication status
-

Required MySQL Permissions

GRANT PROCESS, REPLICATION CLIENT, SELECT ON *.* TO 'exporter'@'%';

FLUSH PRIVILEGES;

Permission Details

PROCESS

Allows viewing server processes.

REPLICATION CLIENT

Allows replication-related metrics.

SELECT

Allows reading system information.

Alertmanager Configuration

Basic Configuration

global:

resolve_timeout: 5m

route:

receiver: default

receivers:

- name: default

resolve_timeout

Defines when an alert is considered resolved.

route

Defines alert routing logic.

receiver

Defines notification destination.

Alert Rules

Rule Structure

groups:

- name: example-alerts

rules:

- alert: ExampleAlert

expr: some_metric > value

for: 1m

labels:

severity: warning

annotations:

summary: Alert Summary

description: Alert Description

MySQL Alert Rules

MySQL Down

- alert: MysqlExporterDown

expr: mysql_up == 0

for: 1m

labels:

severity: critical

annotations:

summary: MySQL is down

description: MySQL exporter cannot connect to MySQL

Purpose:

- Detects database outages.
-

High Connections

- alert: HighMysqlConnections

expr: mysql_global_status_threads_connected > 50

for: 2m

Purpose:

- Detects excessive active connections.
-

Slow Queries

- alert: MysqlSlowQueries

expr: rate(mysql_global_status_slow_queries[5m]) > 5

for: 5m

Purpose:

- Detects poor query performance.
-

Aborted Connections

- alert: MysqlTooManyAbortedConnections

expr: increase(mysql_global_status_aborted_connects[10m]) > 10

for: 2m

Purpose:

- Detects failed connection attempts.
-

Prometheus Down

- alert: PrometheusDown
 expr: up{job="prometheus"} == 0
 for: 1m

Purpose:

- Detects Prometheus availability issues.
-

Exporter Down

- alert: ExporterDown
 expr: up == 0
 for: 1m

Purpose:

- Detects exporter failures.
-

Test Alert

- alert: TestAlert
 expr: vector(1)
 for: 10s

labels:
 severity: test

annotations:
summary: Test Alert

Purpose:

- Validates Prometheus and Alertmanager integration.
-

Common Troubleshooting

MySQL Exporter Error

Error:

Access denied; you need (at least one of) the SUPER, REPLICATION CLIENT privilege(s)

Cause:

- Missing MySQL permissions.

Solution:

- Grant PROCESS, REPLICATION CLIENT, and SELECT privileges.
-

No Rules Found

Cause:

- `alerts.yml` not mounted into Prometheus container.

Required volume:

volumes:

- `./prometheus.yml:/etc/prometheus/prometheus.yml`
 - `./alerts.yml:/etc/prometheus/alerts.yml`
-

Alertmanager Not Receiving Alerts

Check:

Prometheus Rule Loading

rule_files:
- alerts.yml

Alertmanager Configuration

alerting:
 alertmanagers:
 - static_configs:
 - targets:
 - alertmanager:9093

Network Connectivity

Prometheus must be able to reach:

alertmanager:9093

Email Notifications

SMTP Configuration

global:
 smtp_smarthost: 'smtp.gmail.com:587'
 smtp_from: 'yourgmail@gmail.com'
 smtp_auth_username: 'yourgmail@gmail.com'
 smtp_auth_password: 'APP_PASSWORD'

Email Receiver

receivers:
- name: gmail-alert

email_configs:
- to: 'yourgmail@gmail.com'
send_resolved: true

Important URLs

Prometheus

<http://localhost:9091>

Targets

<http://localhost:9091/targets>

Rules

<http://localhost:9091/rules>

Alerts

<http://localhost:9091/alerts>

Alertmanager

<http://localhost:9093>

Grafana

<http://localhost:3030>

Key Learning Outcomes

- Prometheus Architecture
- Prometheus Configuration
- Scrape Targets
- MySQL Exporter
- Alert Rules
- Alertmanager Integration
- Alert Routing
- Email Notifications
- Monitoring Troubleshooting
- End-to-End Monitoring Workflow